

# LLM-TrajExplorer: Interactive "What-If" Scenario Planning for Trajectory Dynamics in Tourist Hotspots

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## Abstract

Managing popular tourist destinations effectively presents significant challenges due to the complex spatio-temporal dynamics of visitor trajectories. Traditional management methods often fall short in addressing issues arising from large-scale human flows, particularly in enhancing visitor experiences and responding to emergencies. This paper introduces LLM-TrajExplorer, an innovative interactive "What-If Scenario Planning" system. Leveraging the advanced reasoning and generation capabilities of Large Language Models (LLMs), combined with fine-tuning and Retrieval-Augmented Generation (RAG), the system dynamically simulates and analyzes individual and collective spatio-temporal trajectories in tourist hotspots such as Gulangyu. LLM-TrajExplorer empowers scenic area managers to make more informed decisions by simulating the potential impacts of diverse management strategies (e.g., personalized route recommendations) and emergency plans (e.g., responses to peak-hour crowd evacuations), thereby enhancing operational efficiency and visitor satisfaction. This paper details LLM-TrajExplorer's architecture, core technologies, and showcases its application through key "scenario planning" demonstrations.

## CCS Concepts

• **Information systems** → **Spatial-temporal systems**; • **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Natural language processing**.

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## Keywords

Large Language Models, Interactive Tourism Management, Spatio-temporal Trajectory Simulation, Tourism Scenario Planning

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## 1 Introduction

The booming development of tourism has brought economic benefits to popular scenic areas, but also triggered a series of severe management challenges. Taking Gulangyu, a UNESCO World Cultural Heritage site, as an example, its unique historical features and natural landscapes attracted 9.22 million visitors in 2024, with daily visitor numbers ranging between 25,000 and 65,000[1]. This high-density human flow not only potentially reduces tourist experience quality but also impacts cultural heritage preservation and local community life. Scenic area managers urgently require effective tools to address complex issues including crowd control, personalized services, and emergency response. Existing management approaches predominantly rely on historical data analysis and static planning, or adopt passive post-event responses, lacking capability for proactive, dynamic simulation of future scenarios[6].

Traditional trajectory prediction models[5] and crowd simulation tools[2] perform well in analyzing historical patterns and general mobility, but show limitations in simulating novel "scenario planning" or incorporating complex human behaviors (particularly in dynamic contexts). Moreover, these tools typically output data or visualizations requiring professional interpretation, rather than directly translating to actionable insights for managers. While some scenario planning tools[3] exist, they may be overly complex, cost-prohibitive, or lack specialized support for fine-grained spatiotemporal dynamics in tourist hotspots.

To address these challenges, this paper proposes LLM-TrajExplorer, a novel interactive system leveraging fine-tuned Large Language

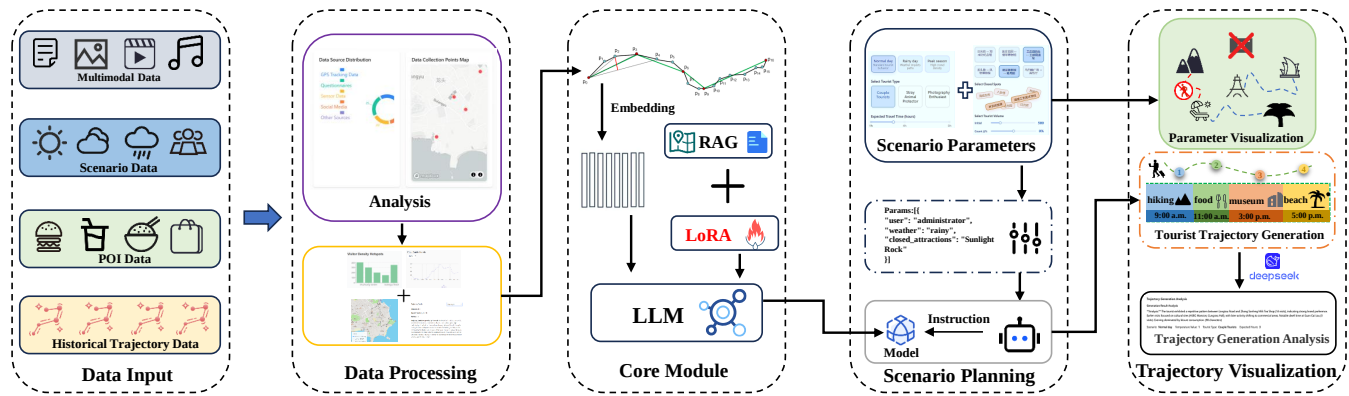


Figure 1: The Five-Layer System Framework of LLM-TrajExplorer

Models (LLMs) combined with Retrieval-Augmented Generation (RAG) technology for "scenario planning" analysis of spatiotemporal trajectories in tourist hotspots. The core innovations of LLM-TrajExplorer include:

- Application of LLMs for predicting and analyzing individual/group-level trajectory dynamics, particularly under hypothetical scenarios
- Interactive interface enabling managers to define scenario parameters through natural language or simple operations, receiving LLM-generated, human-understandable scenario analyses and actionable recommendations
- Context-specific integration of Gulangyu's unique geographical, cultural, and visitor flow data to ensure targeted and practical analysis

LLM-TrajExplorer aims to empower scenic area managers to scientifically evaluate potential impacts of different management strategies and emergency plans, thereby optimizing resource allocation, enhancing visitor experiences, and strengthening emergency response capabilities. The increasingly complex management of tourist hotspots – particularly heritage sites like Gulangyu with dual cultural and natural values – demands more intelligent and adaptive planning tools. Traditional methods based on historical data (which may not apply to new situations or emergencies) struggle with dynamic challenges. Large Language Models bring paradigm shifts to urban computing and scenario planning[9], enabling not just quantitative simulation but also qualitative reasoning, human-like behavior understanding, and natural language explanations, crucial for managing complex human-centric systems like crowds.

## 2 System: LLM-TrajExplorer

The LLM-TrajExplorer system aims to provide scenic area managers with a powerful "what-if" scenario planning platform. Its design integrates large language model technology with tourism-specific optimizations to ensure simulation fidelity and the practicality of recommendations.

### 2.1 System Architecture Overview

The system architecture of LLM-TrajExplorer, depicted in Figure 1, comprises five layers:

- **Input Layer:** Supports scenario definition through interactive maps or parameter forms. It integrates GIS maps, Point of Interest (POI) data, and historical/simulated trajectory data, and provides interfaces for real-time data streams.
- **Data Processing & Knowledge Base Layer:** Maintains a RAG-enhanced knowledge base containing granular information about path features, attraction details, and visitor flow patterns specific to Gulangyu.
- **LLM Core:** Implements trajectory prediction and behavior simulation through fine-tuning and RAG technology, generating Gulangyu-contextualized trajectory dynamics.
- **Scenario Planning Engine:** Translates scenario parameters into LLM instructions and manages simulation workflows.
- **Output & Visualization Layer:** Presents individual/group trajectories, crowd density heatmaps, and key metrics through interactive maps, complemented by LLM-generated natural language analyses.

### 2.2 LLM Core: Fine-Tuning and RAG-Enhanced Spatiotemporal Reasoning

The LLM core deeply integrates fine-tuning and Retrieval-Augmented Generation (RAG) technologies to enable sophisticated spatiotemporal reasoning.

The **LLM Fine-Tuning** process utilizes the DeepSeek-R1-32B[4] model. It transforms multi-source inputs, such as GPS coordinates, behavior logs, and semantic tags, into semantic embeddings to learn universal spatiotemporal patterns[10]. These patterns encompass obstacle avoidance strategies, short-path preferences, and crowd attraction/repulsion dynamics that reflect both physical and psychological behavioral aspects.

**RAG-Enabled Context Awareness** is achieved through a dynamic knowledge base specifically constructed for Gulangyu [7]. This knowledge base provides essential context by incorporating: (1) *Geographic knowledge*, such as road network topology (e.g., "winding paths"), road attributes (width, slope, capacity), and POI details (attraction types, opening hours, attraction popularity metrics); (2) *Behavioral patterns*, derived from historical visitor flow data, holiday crowd distributions, queue times, and preference patterns extracted from tourist reviews via LLM-based sentiment analysis; and (3)

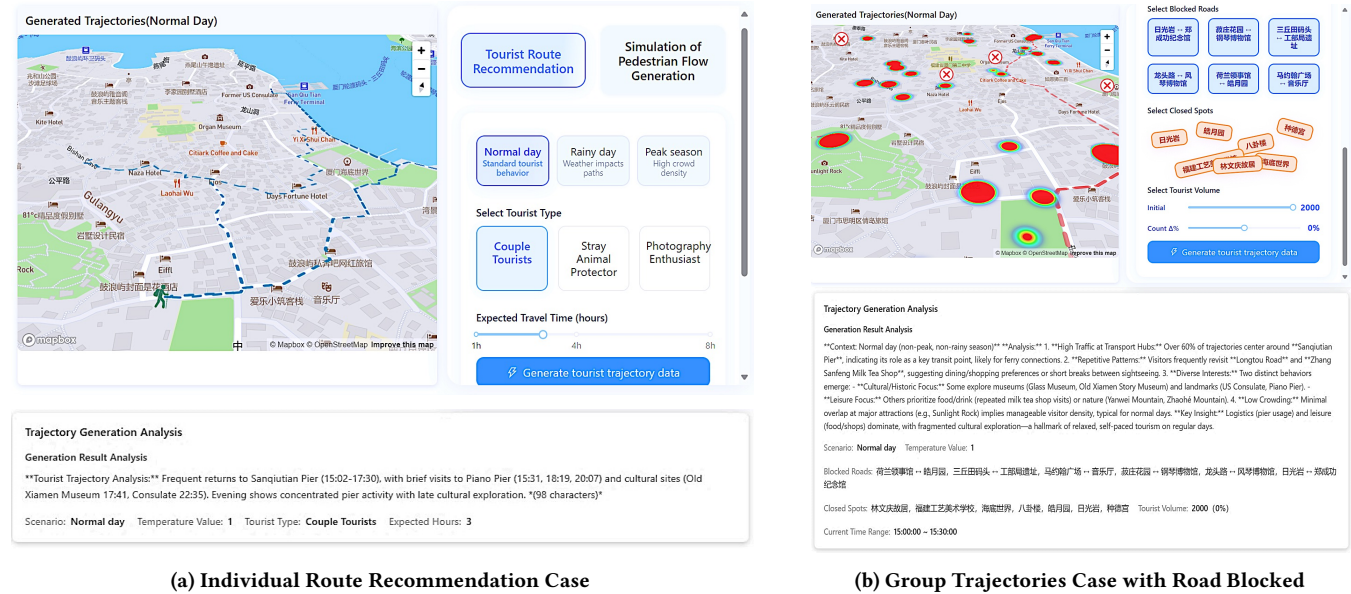


Figure 2: An example of the LLM-TrajExplorer System, showing individual route recommendation and group trajectories under blocked conditions.

*Dynamic information*, like near-real-time data from weather forecasts, ferry schedules, and temporary events, integrated through knowledge-enhanced prompt injection.

This synergy enables the LLM to combine universal behavioral logic (e.g., utility-maximizing behavior) with Gulangyu-specific knowledge (e.g., congestion patterns near stores trending on social media), significantly improving the accuracy of tourist path selection predictions and the value of its decision support capabilities.

## 2.3 Scenario Planning Engine and Trajectory Generation

The Scenario Planning Engine transforms user-defined scenarios into LLM-executable instructions and orchestrates the trajectory generation process.

For **individual trajectory generation**[8], it predicts paths based on visitor profiles (considering type, interests, physical condition, and planned duration), current environmental conditions (such as road closures or attraction shutdowns), and RAG-retrieved contextual information (like alternative routes and the status of nearby attractions).

Subsequently, the engine performs **group dynamics simulation** by aggregating these individual trajectories. This aggregation reveals emergent phenomena such as crowd density fluctuations, identifies congestion levels using various indices, estimates queue lengths, and highlights potential bottlenecks within the scenic area. Building on these simulations, the engine supports **scenario analysis and recommendations**. In this phase, we integrate the simulation outcomes into a structured management task prompt, guiding the LLM to interpret the results (e.g., "Closure of Route A diverts 70% of visitors to narrow Route C, causing a predicted 30-minute congestion") and generates actionable management suggestions

(e.g., "Implement directional flow control at Route C and deploy guidance signs to Route D"). Furthermore, in quantitative comparisons with several advanced baseline models, LLM-TrajExplorer demonstrated optimal performance on multiple key metrics. For instance, its OD similarity (ODSim) metric showed an improvement of over 42% compared to the next-best model, strongly validating the simulation's fidelity.

## 2.4 Interactive Visualization Interface

The interface enables intuitive system operation and interaction through three primary modules, designed to provide comprehensive oversight and control. The **Map Display Module** is central to the interface, featuring an interactive Gulangyu map overlay. This overlay dynamically shows individual trajectories (e.g., represented as color-coded lines) and visualizes collective crowd flows using techniques such as heatmaps, particle flows, or directional arrows. Complementing the map, the **Scenario Control Panel** offers intuitive controls that allow managers to directly manipulate simulation parameters. These include capabilities for map-based road closure simulation, selection of different crowd evacuation modes, and customization of visitor profiles for targeted analysis. Finally, the **Analysis & Recommendation Panel** displays real-time, LLM-generated outputs. This includes textual scenario analyses, key performance indicators (KPIs) like estimated transit times and congestion indices, and specific management recommendations tailored to the simulated conditions.

The system emphasizes immediate visual feedback and natural language explanations to bridge the gap between complex simulations and operational decision-making.

### 3 Demonstration

LLM-TrajExplorer is demonstrated through a web application interface that showcases its interactive capabilities in assisting tourists with personalized itinerary planning and supporting scenic area managers with preliminary “what-if” scenario planning. The current demonstration version of LLM-TrajExplorer mainly includes two core modules.

The first module, focused on personalized trajectory generation for tourists, aims to enhance their travel experience by facilitating personalized service planning. Regarding user interaction and system response: tourists can select current or expected weather conditions, set personal preferences (e.g., interest in encountering stray animals), and specify their visit duration. Based on these inputs, the Large Language Model (LLM) generates multiple recommended tour routes. The system then provides a preliminary analysis for each route. On the map, tourists can observe simulated individuals moving according to the selected trajectory and can easily switch between alternative travel plans by clicking on different trajectory indices. This module allows tourists to explore scenarios such as, “What itinerary options do I have if I choose these preferences and conditions?” as shown in Figure 2a, thereby embodying the “what-if” scenario planning concept.

The second module is designed for managers, emphasizing crowd flow simulation and preliminary emergency observation. This module assists scenic area managers in more intuitively observing visitor flow under different operational conditions, particularly during emergencies, to support decision-making. Its user interaction and system response process is as follows: Managers first generate a baseline simulated crowd flow based on selected preset scenarios (e.g., crowd benchmarks for weekdays or holidays). They can then simulate various ‘unexpected’ situations, such as the closure of a specific attraction or a road blockage between two attractions. The system subsequently simulates and dynamically displays the resulting changes in visitor flow. By observing these changes, especially when simulations indicate high crowd density that might necessitate evacuation, managers can better evaluate the situation and receive preliminary guidance for developing response strategies, such as crowd redirection and evacuation route selection. This module helps managers explore scenarios like, “How will crowd flow change if this attraction is closed or this road is impassable?” as shown in Figure 2b, facilitating preparation for operational adjustments and emergency responses.

Through these two core modules, LLM-TrajExplorer demonstrates its application potential. For tourists, it leverages the LLM to understand diverse preferences and generate personalized suggestions aimed at enhancing satisfaction. For managers, it provides a “what-if” scenario platform to simulate events like attraction closures, observe dynamic crowd responses, and offer intuitive decision support for both daily operations and emergency management, such as crowd dispersal in high-density situations.

### 4 Future Work

In the future, we will further optimize the inference speed of LLM to support larger scale, more real-time trajectory simulations, and integrate more refined psychological and social behavior models into LLM-driven agents. In addition, technologies such as model

compression and knowledge distillation will be explored to prepare for the deployment of lightweight models on edge devices or mobile terminals of management personnel in scenic areas. Conduct in-depth usability testing and interviews with actual users such as scenic area management personnel and emergency response teams on Gulangyu Island, collect their needs and feedback on the system interface, interaction methods, analysis results presentation, and suggested content, and continuously optimize the system based on this to ensure that it truly meets the actual workflow and decision-making needs. Pending cooperation with other scenic areas, we aim to expand the application scenarios to additional popular tourist destinations, thereby improving the system’s robustness.

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